Modeling Techniques FAQ's

Q1: Should foreign keys be displayed in the diagrams?
A1: No, neither in ER nor UML.

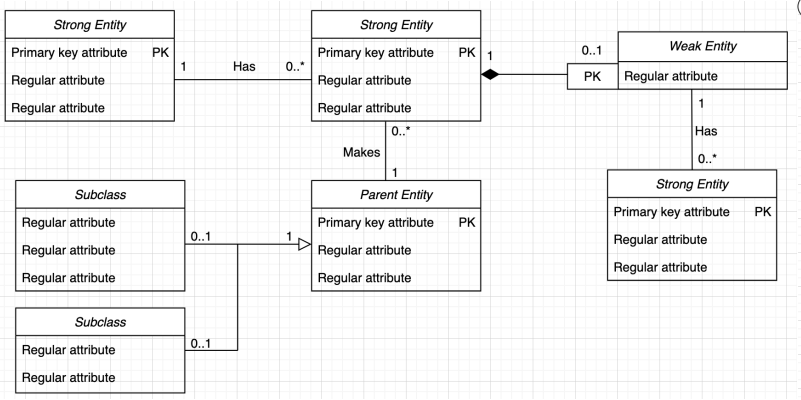
Q2: Can an entity be connected to multiple other entities?
A2: Yes, if the entity has multiple foreign keys it should be connected to multiple entities.

Q3: What is a weak entity?
A3: An entity that requires a foreign key to form a primary key.

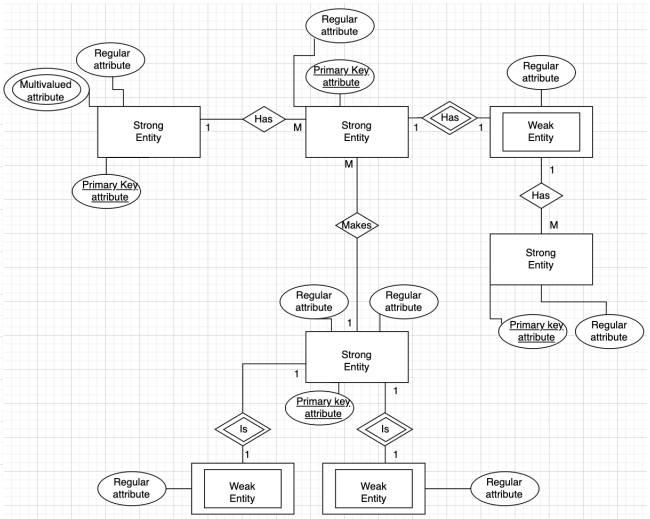
Q4: Let's say we have a weak entity with primary key "a" and another entity with a primary key which isn't a foreign key. Furthermore, the second entity has an attribute which is a foreign key referencing "a". Should the second entity have a weak relationship with the weak entity because the second entity has a foreign key referencing the very attribute which makes the first entity weak?
A4: No, even though the second entity has a foreign key referencing the weak entity's primary key, the second entity does not have a weak relationship with the weak entity because it does not have a foreign key as its primary key.

Q5: What is the difference between cardinality and multiplicity?
A5: Cardinality is used for ER and can either be 1-1, 1-M or M-N. Multiplicity is used for UML diagrams and can either be nr–nr, nr..nr–nr, nr..nr–nr..nr, nr-* or nr..nr-* where nr is an integer, nr..nr represents an interval, and * means many.

Q6: What is an example of how an ER and UML diagram should look like?
A6: Below is an example of an ER diagram and a UML diagram. Subclass, Parent Entity, Strong Entity and Weak Entity should all be replaced with names of the entities. The same goes for the attributes.



UML



ER

Q7: How should multivalued attributes be represented?
A7: In ER, make a double oval. In UML, represent it like a regular attribute.

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Q8: How should composite primary keys be represented if none of the primary key attributes are foreign keys?
A8: If none of the primary key attributes are foreign keys, the entity is strong. In UML, use the PK notation next to all attributes in the primary key (excluding any foreign key since those are not shown at all). In ER, underline each attribute of the primary key, again excluding foreign keys since those are not present.

Q9: How should composite primary keys be represented if some of the primary key attributes are foreign keys?
A9: This is a weak entity. In UML, treat it like a normal weak entity (including the black rhombus arrow and the PK-box) but add the PK notation next to each primary key attribute which isn't a foreign key. In ER, treat it like a normal weak entity but use a dashed underline for each attribute which is part of the composite primary key but isn't a foreign key.

Q10: What is a composition in UML?
A10: Composition is used when one entity cannot exist in a meaningful way without the other. The composition arrow is also used together with the PK-box for weak relationships. An important rule is that the end of the black diamond-arrow must have multiplicity 1 (which may also be written as 1..1).

Q11: What is an aggregation in UML?
A11: Aggregation is used when an entity can exist independently in a meaningful way of another entity but still belongs to it. Let's say we have an entity called "Wheel" and another called "Car". If Wheel has a primary key "wheel_ID", it can exist independently of Car. Furthermore, if we have more vehicle entities such as "Bike" and "Bus", Wheel can have aggregations to those as well. For a relationship to be an aggregation and not composition, it is not enough that the child can exist independently of the parent, it must also be able to exist in a meaningful way without its parent. A wheel might exist in a meaningful way without a car because it can be part of another vehicle or because we may want to have a record of all wheels, even those not being used. A rule is that at the end of the white diamond-arrow, the multiplicity label must be 0..1.

Q12: What is inheritance/subclasses in UML?
A12: Use the subclass notation (white triangle-arrow) when a weak entity is a type of its parent entity. For example, if we have entities "SportsCar", "Truck" and "Car", where "SportsCar" and "Truck" have primary key "carID" which is a foreign key referencing Car, then SportsCar and Truck should be subclasses of Car.

Q13: How do we handle junction tables?
A13: In both ER and UML, put the junction table name as the relationship type (for example, "hasFood" instead of the generic "has" if there is a junction table connecting hotels with types of food). In UML, use association between two tables which are connected using a junction table.

Q14: In UML, when should relationships be named?
A14: Only for associations, not for aggregations, compositions or inheritance.